Supplementary material for: A unified two-dimensional risk quantification framework for highway work zones considering intensive multi-vehicle interactions

Jiaxin Li[a,b], Hao Cheng[c], Yu Zhou[d] and Min Zhang[a]

[a] School of Transportation Engineering, Chang'an University, Xi'an, China
[b] Hangzhou International Innovation Institute, Beihang University, Hangzhou, China
[c] School of Vehicle and Mobility, Tsinghua University, Beijing, China
[d] School of Transportation Science and Engineering, Beihang University, Beijing, China

Guide to the supplement

This document accompanies the main article and provides extended geometric formulations, numerical procedures, evaluation details, and additional case studies. Sections, equations, figures, tables, and algorithms use the S prefix.

- **Section S1:** Calculation of anticipated leaving time for parallel and non-parallel conflict scenarios.
- **Section S2:** Numerical computation of CEDI for MLC conflicts, including search settings and serial computational efficiency.
- **Section S3:** Numerical computation of vehicle-specific CEDI for vehicle conflicts, including search settings and serial computational efficiency.
- **Section S4:** Monte Carlo simulation settings and evaluation of RCD detection performance.
- **Section S5:** Additional MLC and vehicle-conflict case studies illustrating early warning and risk asymmetry.
- **Section S6:** Geometric explanation of abrupt changes in ACT for MLC conflicts.

S1. Calculation of anticipated leaving time in two scenarios

This section details the calculation of ALT_C for parallel and non-parallel conflicts, classified by the travel directions of vehicles A and B. A parallel conflict occurs when the vehicles travel in the same direction, whereas a non-parallel conflict occurs when their travel directions differ. **Fig. S1** illustrates the conflict paths in these two scenarios. The general formulation given in the main article is

$$ALT_C = \frac{D_C}{\left\| \boldsymbol{v}_{CB} \times \boldsymbol{\theta}_{AB} \right\|}. \tag{S1}$$

The calculation of the anticipated leaving time of vehicle C depends on the corner intersection pairs retained at the shortest conflict distance. These pairs may define either a single conflict path or two distinct paths, corresponding to different conflict scenarios. Accordingly, the leaving distance D_C is determined separately for each scenario based on the associated vertex geometry, as detailed below.

Non-parallel conflict scenarios

Fig. S1(a) depicts a non-parallel conflict scenario, in which the selected conflict corners define a single conflict path. The leaving distance D_C is therefore determined by the displacement required for the entire body of vehicle C to clear this line. Accordingly, the signed projection in Eq. (14) of the main article reduces to

$$d_{VC} = \overline{V_i C} \times \boldsymbol{\theta}_{AB}. \tag{S2}$$

Accordingly, Eq. (15) of the main article simplifies to

$$D_C = \max\left\{0,\ d_C - \operatorname{sgn}\left(\boldsymbol{v}_{CB} \times \boldsymbol{\theta}_{AB}\right) d_{VC}\right\}. \tag{S3}$$

Parallel conflict scenarios

Fig. S1(b) illustrates a parallel conflict scenario, in which the selected corner intersection pairs define two distinct conflict lines, indicating that the set of selected collision vertices contains more than one element. Unlike the single-line configuration, evaluating spatial obstruction persistence requires determining the time needed for vehicle C to completely clear both conflict lines. Accordingly, the leaving distance D_C is determined by the last conflict line cleared in the direction of the relative motion of vehicle C. The corresponding calculations follow Eqs. (14) and (15) in the main article.

We emphasize that perfectly parallel vehicle trajectories are rarely observed in real-world traffic. Nevertheless, conflicts involving vehicles with sufficiently small differences in their travel directions can be approximated as parallel conflicts, particularly in car-following situations in highway work zones.

S2. CEDI computation for MLC conflicts

S2.1. Numerical solution

This section summarizes the numerical computation of CEDI for an RCD-identified MLC conflict. The notation and normalization scales follow Section 3.3.2 of the main article. A candidate strategy is specified by (a_x, τ), with lateral avoidance initiated at $t + \tau$ and $\boldsymbol{u} = (a_x, a_y(\tau))$. Equation numbers (19)–(33) refer to the main article.

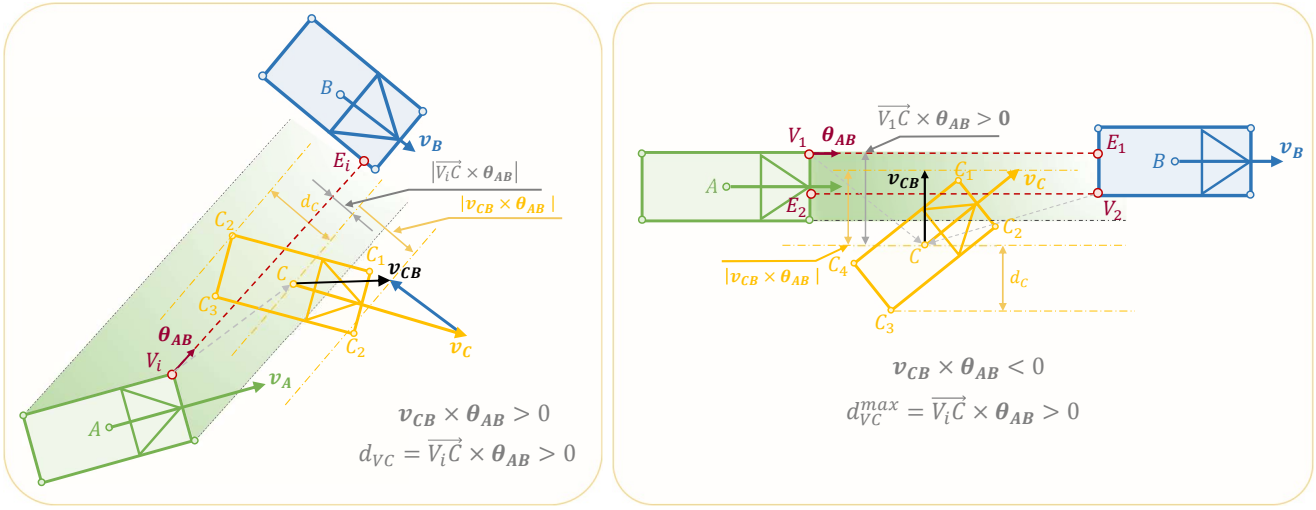


Figure S1: Calculation of the anticipated leaving time of vehicle C in (a) non-parallel and (b) parallel conflict scenarios.

Step 1: Generate candidate strategies.

For each candidate longitudinal acceleration a_x, calculate the first work zone collision time $T(a_x)$ using Eq. (30). Waiting durations must satisfy Eq. (31):

$$0 \le \tau < T(a_x), \qquad v(t) + a_x \tau > 0. \tag{S4}$$

The numerical search range and increments are reported below. They specify the evaluated candidates without changing the physical constraints in the main article.

Step 2: Predict states and check clearances.

Identify the nearest preceding and following vehicles in the original lane. For each candidate (a_x, τ), predict the SV and surrounding states at $t+\tau$ and determine $\mathcal{V}_t(\tau)$. Calculate $b_{o,r}(\tau)$ using Eqs. (25)–(26); at $\tau = 0$, set $b_{o,r}(0) = 0$ because the waiting interval vanishes. Check the original-lane clearances during the waiting interval and all relevant target-lane clearances at LC initiation using Eq. (32). Exclude candidates violating these conditions or the road-boundary constraints specified following Eq. (19). For a retained gap, identify $f(\tau)$ and $r(\tau)$.

Step 3: Calculate evasive and interaction demands.

Estimate $ET(\tau)$ and InDepth(τ) at LC initiation and, for $ET(\tau) > 0$, calculate $a_y(\tau)$ using Eq. (22). $T(a_x)$ limits the waiting duration, whereas $ET(\tau)$ determines the remaining time for lateral avoidance. Calculate $b_{t,f}(\tau)$ and $b_{t,r}(\tau)$ using Eqs. (23)–(24), and $b_{o,f}(\tau)$ using Eq. (28). An absent neighbor contributes zero demand. Following Eqs. (27) and (29), the aggregate demands are

$$\begin{aligned} a_r^s(\tau) &= b_{o,r}(\tau) + b_{t,r}(\tau), \\ a_f^s(\tau) &= b_{o,f}(\tau) + b_{t,f}(\tau). \end{aligned} \tag{S5}$$

Step 4: Evaluate the cost and retain the best candidate.

Evaluate $C(\boldsymbol{u})$ using Eq. (21) for each retained candidate, as required by the MLC formulation in Eq. (33). Let C_{best} store the lowest cost among the feasible candidates evaluated so far. Initialize it to $+\infty$ and update it as

$$C_{\text{best}} \leftarrow \min\{C_{\text{best}}, C(\boldsymbol{u})\}. \tag{S6}$$

Thus, C_{best} is an auxiliary algorithm variable; the definition of CEDI as the minimum feasible maneuver cost is unchanged. The returned value numerically approximates CEDI over the specified search range and resolution. If $\mathcal{U}_m = \varnothing$, the theoretical CEDI is $+\infty$; a finite search returning $C_{\text{best}} = +\infty$ indicates only that no evaluated candidate was feasible.

Numerical settings and computational acceleration.

Candidate longitudinal accelerations are sampled over the reported search interval with increment Δa_x, and waiting durations are sampled with increment $\Delta\tau$. The search-grid endpoints are included, and zero is added explicitly if it is not already a grid point. Waiting durations equal to $T(a_x)$ are excluded according to Eq. (31). Because $T(a_x)$ may be $+\infty$, a finite numerical waiting-time limit $\tau_{\max}$ is specified. Evaluated waiting durations therefore satisfy both $0 \le \tau \le \tau_{\max}$ and Eq. (31). This limit restricts the numerical search rather than redefining $\mathcal{U}_m$; it is not automatically set equal to the initial $ET(t)$. **Table S1** summarizes the numerical search settings, which are distinct from the normalization scales defined in the main article. Each search parameter was determined through iterative convergence tests, ensuring that the difference between successive optimal objective values was below 0.01.

The candidate evaluation avoids redundant work by identifying the original-lane neighbors once at each observation time and calculating $T(a_x)$ once for each candidate a_x. Clearance checks precede evaluation of the complete cost. The longitudinal component of the cost is used for lower-bound screening, as shown in **Algorithm S1**.

Table S1
Search settings and normalization scales for MLC CEDI computation.

Setting	Value
Longitudinal acceleration search interval	$[-0.75g,\ 0.20g]$
Number of longitudinal acceleration samples	30
Number of waiting-time samples	30
$a_{x,\mathrm{lim}}(a_x)$ for $a_x > 0$	$0.20g$
$a_{x,\mathrm{lim}}(a_x)$ for $a_x \le 0$	$0.75g$
$a_{y,\max}$	$0.30g$
$a^s_{f,\max}$ and $a^s_{r,\max}$	$0.75g$ for both

Algorithm S1 CEDI computation for MLC conflicts

Input: SV and surrounding vehicle states, work zone geometry, road boundaries, and search settings
Output: Numerical estimate of CEDI(t)
Initialize $C_{\mathrm{best}} \leftarrow +\infty$
Identify the nearest original-lane preceding and following vehicles
for each candidate longitudinal acceleration a_x **do**
  if $|a_x|/a_{x,\mathrm{lim}}(a_x) \ge C_{\mathrm{best}}$ **then**
    continue
  end if
  Calculate the first work zone contact time $T(a_x)$
  for each candidate $\tau \in [0, \tau_{\max}]$ satisfying the waiting-time constraints **do**
    Predict states at $t + \tau$ and determine $\mathcal{V}_t(\tau)$
    Calculate $b_{o,r}(\tau)$, with $b_{o,r}(0) = 0$
    if the clearance and road-boundary constraints are satisfied **then**
      Identify $f(\tau)$ and $r(\tau)$; estimate $ET(\tau)$ and InDepth(τ)
      if $ET(\tau) > 0$ **then**
        Calculate $a_y(\tau)$ and the individual interaction demands
        Form $a^s_f(\tau)$ and $a^s_r(\tau)$ using **Eq. (S5)**
        Set $\boldsymbol{u} = (a_x, a_y(\tau))$ and evaluate the common avoidance cost $C(\boldsymbol{u})$
        $C_{\mathrm{best}} \leftarrow \min\{C_{\mathrm{best}}, C(\boldsymbol{u})\}$
      end if
    end if
  end for
end for
return C_{best} as the numerical CEDI estimate

S2.2. Serial computational efficiency

Implementation and computing environment.

The implementation uses Python 3.13, NumPy 2.5.3, and pandas 3.0.5. The serial benchmark runs on an Intel Core i5-14600K CPU with 32 GB of RAM under Windows 11, using one process and one computational thread without GPU acceleration. Computation time is measured using `time.perf_counter()` for one vehicle–work-zone pair at one observation time. It includes neighbor identification, state prediction, candidate search, demand calculation, feasibility checks, and minimum-cost selection, excluding file input/output.

Across the three datasets, CEDI evaluation for MLC conflicts requires a mean computation time of 1.95 ms/frame. The 90th, 95th, and 99th percentile computation times are 3.64, 4.81, and 8.02 ms/frame, respectively. These results indicate that CEDI can support assessment of MLC conflicts at a low computational cost under the tested conditions.

This benchmark uses the parameter settings reported in **Table S1**. The reported computation times apply specifically to these settings and the stated computational environment. Expanding the parameter search ranges or increasing the sampling density generally increases computational cost.

S3. CEDI computation for vehicle conflicts

S3.1. Numerical solution

This section summarizes the numerical computation of CEDI for an RCD-identified vehicle conflict. The notation and normalization scales follow Section 3.3.2 of the main article. Each interacting vehicle is evaluated separately as the SV. For one evaluation, let s denote the SV, p the other interacting vehicle, and $\mathcal{V}_s$ the remaining surrounding vehicles. The partner p is included in the collision constraints and excluded from the surrounding-vehicle demand sums. The procedure uses the common avoidance-cost function defined in the main article, with preceding- and following-vehicle demands aggregated separately.

Step 1: Generate candidate strategies.
Calculate the initial ET of the original interacting pair using the geometric contact-time formulation in the main article. For a conflict with $ET > 0$, retain this interval for every candidate $\boldsymbol{u} = (a_x, a_y)$. Candidate controls include longitudinal acceleration or braking, steering in either direction, and their combinations. The numerical search range and resolution are reported below.

Step 2: Predict states and identify potential interactions.
Predict the SV trajectory over $0 \le \xi \le ET$ using a_x along its initial motion direction and the symmetric lateral maneuver described in the main article. The other interacting vehicle retains its current velocity. For each $j \in \mathcal{V}_s$, identify any first sampled conflict trigger $t_j(\boldsymbol{u})$ using the pairwise geometric prediction. When a trigger exists, classify j as preceding or following according to its predicted longitudinal position relative to the SV at that time. The trigger indicates a potential conflict; it is not the actual collision time.

Step 3: Calculate interaction demands and check feasibility.
For a following vehicle, calculate $b_j^r(\boldsymbol{u})$ as the larger of the braking demands needed during the maneuver and after its completion. Both demands refer to the same response beginning at $t_j(\boldsymbol{u})$; braking is not restarted at ET. A preceding vehicle maintains its current velocity. If a subsequent conflict is predicted from the terminal states, $b_j^f(\boldsymbol{u})$ represents the SV's subsequent braking demand. A vehicle that requires no response contributes zero demand.

Following the definitions in the main article, the aggregate demands are

$$\begin{aligned} a_f^s(\boldsymbol{u}) &= \sum_{j \in \mathcal{V}_f(\boldsymbol{u})} b_j^f(\boldsymbol{u}), \\ a_r^s(\boldsymbol{u}) &= \sum_{j \in \mathcal{V}_r(\boldsymbol{u})} b_j^r(\boldsymbol{u}). \end{aligned} \tag{S7}$$

Individual following-vehicle demands determine their braking trajectories; the aggregate demands enter the avoidance cost.

Examine the safety constraints of the vehicle-conflict formulation over the maneuver interval. The SV must avoid p and all surrounding vehicles. The SV and the following vehicles that actually brake must also avoid $\mathcal{W}$ and remain within Ω. These checks use the individual response trajectories rather than the aggregate demands. A contact with a preceding vehicle during the maneuver causes candidate rejection.

Geometric feasibility is checked over the piecewise trajectories using SAT, separately from the sampled trigger queries. At $t + ET$, extrapolate the original interacting pair under constant velocity and reject the candidate if a subsequent contact is predicted.

Step 4: Evaluate the cost and retain the best candidate.
For each feasible candidate, substitute a_x, a_y, $a_f^s(\boldsymbol{u})$, and $a_r^s(\boldsymbol{u})$ into the common avoidance-cost function in Section 3.3.2 of the main article. The preceding and following demands are each summed before their normalized values are squared. Let C_{best} store the lowest cost among the feasible candidates evaluated so far. Initialize it to $+\infty$ and update it as

$$C_{\text{best}} \leftarrow \min\{C_{\text{best}}, C(\boldsymbol{u})\}. \tag{S8}$$

The returned value numerically approximates CEDI over the specified candidate set. If the continuous feasible set $\mathcal{U}_v$ is empty, the theoretical CEDI is $+\infty$. A finite search returning $C_{\text{best}} = +\infty$ indicates only that no evaluated candidate was feasible. Repeat the procedure with the two interacting vehicles exchanging roles to obtain their separate CEDI values.

Numerical settings and computational acceleration.
Table S2 summarizes the numerical search settings for vehicle conflicts. Zero longitudinal and lateral accelerations are included in the candidate set, and the trigger sampling includes both the initial instant and ET. These numerical settings are distinct from the physical constraints and the cost normalization scales.

Previously sampled controls are retained across search expansions. The final longitudinal and lateral search intervals are $[-3.00g, 0.80g]$ and $[-1.20g, 1.20g]$, respectively, while the cost normalization scales remain unchanged. The axis-grid parameters control supplementary sampling near the coordinate axes rather than a full 30×30 grid. The normalization follows the main article: $a_{x,\text{lim}}(a_x) = 0.20g$ for $a_x > 0$ and $0.75g$ otherwise, $a_{y,\max} = 0.30g$, and $a_{f,\max}^s = a_{r,\max}^s = 0.75g$. To reduce repeated calculations, the initial ET and reusable geometric quantities are computed once for each SV evaluation. Candidates are ordered by the nondecreasing SV-only cost lower bound

$$L(\boldsymbol{u}) = \sqrt{\left(\frac{a_x}{a_{x,\text{lim}}(a_x)}\right)^2 + \left(\frac{a_y}{a_{y,\max}}\right)^2} \le C(\boldsymbol{u}). \tag{S9}$$

When $L(\boldsymbol{u}) \ge C_{\text{best}}$, the candidate and all subsequent candidates in this order cannot improve the incumbent. Infeasible candidates are rejected before the complete cost is evaluated. If partial-cost screening is used, it combines the SV terms with the accumulated preceding-vehicle and following-vehicle demands, with each group sum normalized and squared. This preserves the aggregate-demand definition in **Eq. (S7)**. **Algorithm S2** summarizes the procedure.

S3.2. Serial computational efficiency

The implementation uses Python 3.13, with NumPy 2.5.3 for numerical operations, pandas 3.0.5 for data organization, and Numba for JIT compilation. The serial benchmark is performed on an Intel Core i5-14600K CPU with 32 GB of RAM running Windows 11, using one process and one worker without GPU acceleration. JIT compilation is completed before timing. Computation time is measured using `time.perf_counter()` for one SV perspective of a vehicle pair at one observation time, including geometric preparation, candidate search, response-demand calculation, feasibility checks, and minimum-cost selection. File input/output and plotting are excluded.

Table S2
Numerical search settings for vehicle-conflict CEDI computation.

Setting	Value
Initial longitudinal acceleration search interval	$[-0.75g,\ 0.20g]$
Initial lateral acceleration search interval	$[-0.30g,\ 0.30g]$
Candidate sampling method	Sector-based sampling
Normalized radial sampling increment	$1/60$ per level
Axis-grid parameters `ax_steps` and `ay_steps`	30 and 30
First-trigger sampling interval	0.1 s
Search expansion factors	1, 1.5, 2, 3, 4

Table S3
Definitions and value ranges of simulation parameters.

Parameter	Definition	Range
$v_{i,x}^0$	Initial longitudinal speed of vehicle i (m/s)	$[10, 25]$
$v_{i,y}^0$	Initial lateral speed of vehicle i (m/s)	$[-1.0, 1.0]$
x_i^0	Initial longitudinal position of vehicle i (m)	$[0, 50]$
y_i^0	Initial lateral position of vehicle i (m)	$[0, 11.25]$
$a_{i,x}^t$	Longitudinal acceleration of vehicle i (m/s^2)	$[-3.0, 3.0]$
$a_{i,y}^t$	Lateral acceleration of vehicle i (m/s^2)	$[-1.0, 1.0]$
$v_{i,x}^t$	Longitudinal speed of vehicle i at time t (m/s)	$[5, 30]$
$v_{i,y}^t$	Lateral speed of vehicle i at time t (m/s)	$[-1.5, 1.5]$
y_i^t	Lateral position of vehicle i at time t (m)	$[0, 11.25]$
L_i	Length of vehicle i (m)	$[4, 14]$
w_i	Width of vehicle i (m)	$[1.7, 2.7]$

Algorithm S2 CEDI computation for vehicle conflicts

Input: SV, interacting and surrounding vehicle states, work zone geometry, road boundaries, and search settings
Output: Numerical estimate of CEDI(t) for the SV
Initialize $C_{\text{best}} \leftarrow +\infty$
Calculate the initial ET and precompute geometric quantities
Order candidate controls by nondecreasing $L(\boldsymbol{u})$
for each candidate $\boldsymbol{u} = (a_x, a_y)$ **do**
  if $L(\boldsymbol{u}) \geq C_{\text{best}}$ **then**
    break
  end if
  Predict trajectories over $[0, ET]$, keeping the interacting vehicle at constant velocity
  Identify triggered surrounding vehicles and their roles
  Calculate individual response demands, assigning zero when no response is required
  if all required individual responses are admissible **then**
    if all safety and terminal conditions are satisfied **then**
      Calculate $a_f^s(\boldsymbol{u})$ and $a_r^s(\boldsymbol{u})$ using **Eq. (S7)**
      Evaluate the common avoidance cost $C(\boldsymbol{u})$
      $C_{\text{best}} \leftarrow \min\{C_{\text{best}}, C(\boldsymbol{u})\}$
    end if
  end if
end for
return C_{best} as the numerical CEDI estimate

Across the three datasets, CEDI evaluation for vehicle conflicts requires a mean computation time of 4.71 ms/frame. The 90th, 95th, and 99th percentile computation times are 7.43, 10.89, and 26.20 ms/frame, respectively. Each record corresponds to one evaluation from the SV perspective of a vehicle pair at one observation time.

Computational demand depends on the number of surrounding vehicles that must be screened for potential interactions. For each candidate maneuver, these vehicles require additional conflict-trigger queries, response-demand calculations, and safety checks. Lane closures in highway work zones concentrate interactions within limited road space,

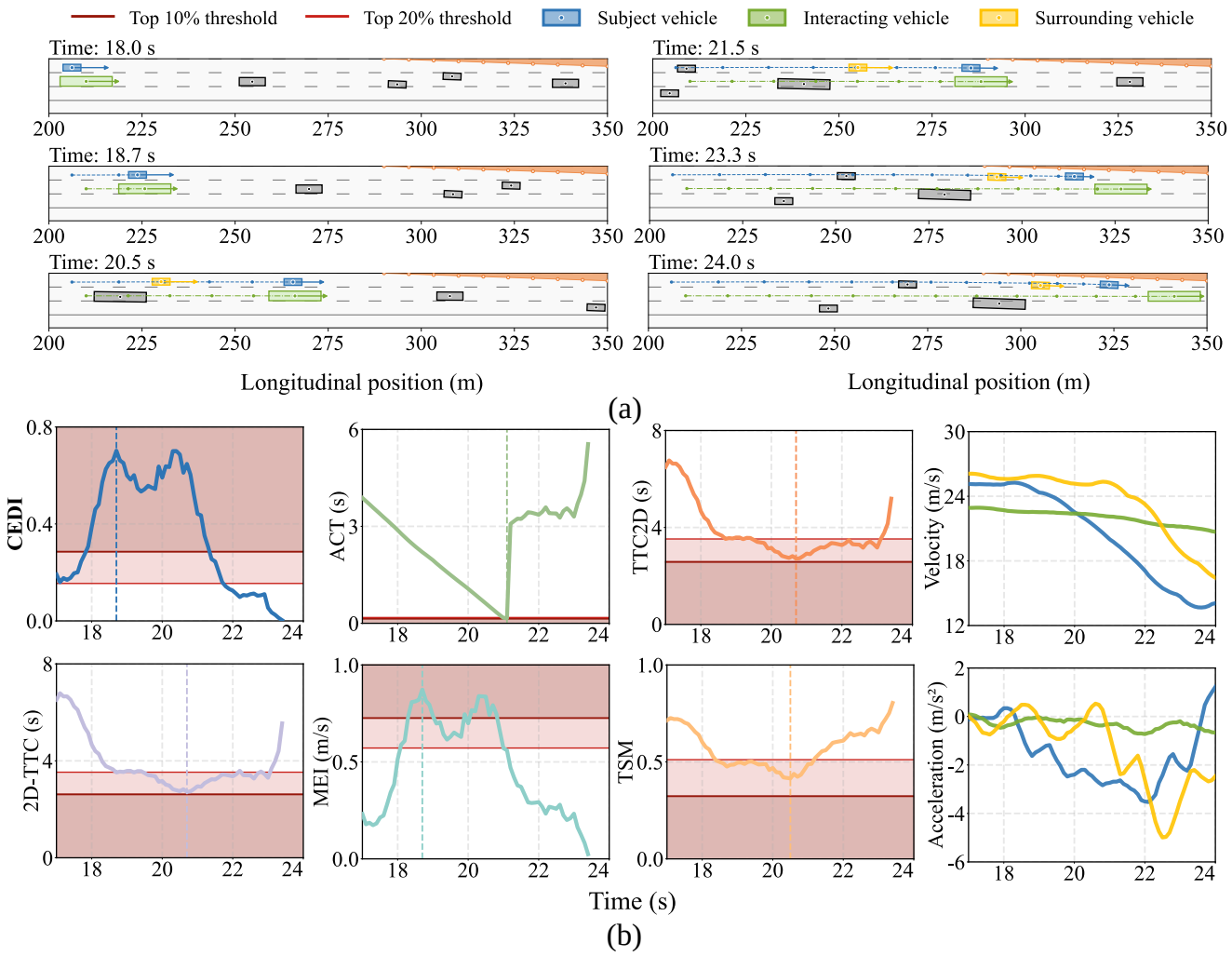


Figure S2: Additional Case S1: An aborted attempt by the SV to overtake the interacting vehicle. (a) Vehicle configurations at six selected instants. Arrows originate at vehicle centers and indicate travel directions, with lengths proportional to vehicle speeds. (b) Temporal evolution of six SSMs and the vehicle speed and acceleration profiles.

making the computational workload dependent on the surrounding traffic configuration. The reported times therefore apply specifically to the parameter settings in **Table S2** and the tested traffic conditions. Expanding the search ranges or increasing the sampling density generally increases computational cost.

S4. Monte Carlo simulation for evaluating RCD performance

This section outlines the simulation parameters used in the main article for evaluating RCD performance based on Monte Carlo-simulated collision data, and examines RCD missed-detection scenarios within multi-vehicle environments.

S4.1. Scenario generation and parameter settings

In this study, empirical trajectory data collected from work zones are utilized to calibrate the simulation parameters. The specific parameter values and their sampling ranges are summarized in **Table S3**.

S4.2. RCD performance in multi-vehicle scenarios

Due to the limited availability of real-world collision trajectory data, Monte Carlo simulations were conducted to assess missed detections by RCD in multi-vehicle scenarios. 1,000,000 valid initial configurations were generated for two-, three-, four- and five-vehicle settings, with initially overlapping vehicle bodies excluded. Each configuration was evaluated under two motion settings: constant velocity and time-varying longitudinal and lateral acceleration. The parameter ranges are specified in **Table S3**, including the initial positions and velocities, acceleration and velocity bounds, vehicle dimensions, and roadway width. An independent geometric contact calculation identified the first physical contact in each simulation, at which point the simulation was terminated. RCD predictions before that contact were then examined. For configurations with at least three vehicles, cases in which another vehicle satisfied the spatial screening conditions and caused the predicted conflict to be excluded as unreachable were recorded separately.

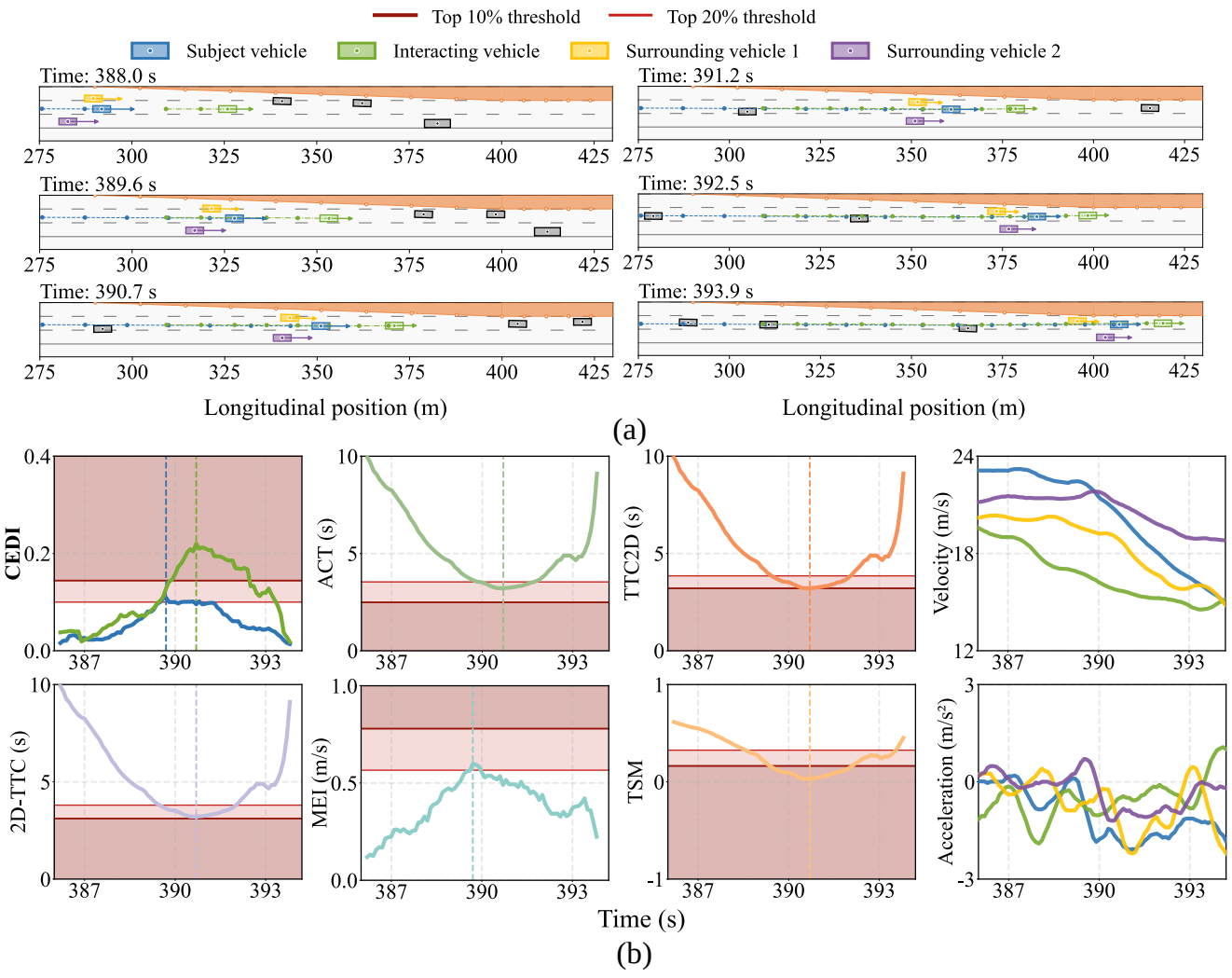


Figure S3: Additional Case S2: A vehicle conflict under surrounding traffic constraints. (a) Vehicle configurations at six selected instants. Arrows originate at vehicle centers and indicate travel directions, with lengths proportional to vehicle speeds. (b) Temporal evolution of six SSMs and the vehicle speed and acceleration profiles. CEDI is evaluated separately for the SV and the interacting vehicle.

As shown in **Table S4**, RCD achieved a detection rate of 100% under constant-velocity motion and approximately 99.99% under time-varying acceleration. The detection rate was slightly lower under time-varying acceleration because RCD assumes a constant-velocity model. Together with the reduction in candidate conflicts reported in the main article, these results indicate that RCD can reduce the events passed to subsequent risk quantification while maintaining a high detection rate in the evaluated simulations.

Table S4
Missed collisions under different motion settings.

Motion setting	Detection rate	Missed collisions
CV	100%	0
Time-varying kinematic model	99.99%	35

S5. Additional case studies

This section presents additional MLC and vehicle-conflict case studies that complement the main article.

Additional case S1: Failed overtaking before MLC

Fig. S2 illustrates an MLC conflict in which the SV must merge out of the closing lane while an interacting vehicle occupies the adjacent lane. At $t = 18.0$ s, the SV is still relatively far upstream of the work zone and travels alongside the interacting vehicle. Nevertheless, CEDI already exceeds its top 20% risk threshold, reflecting the difficulty of merging under the surrounding traffic constraints. At $t = 18.7$ s, the SV is decelerating, and both CEDI and MEI exhibit pronounced peaks. CEDI reaches a second peak around $t = 20.5$ s, whereas TTC2D, 2D-TTC, and TSM reach their minima approximately between $t = 20.5$ s and $t = 21.0$ s. As the SV continues to decelerate, the interacting vehicle moves ahead, opening a gap for the SV to merge behind it. CEDI decreases markedly as this gap develops. By approximately $t = 23.3$ s, the SV has fallen behind the interacting vehicle

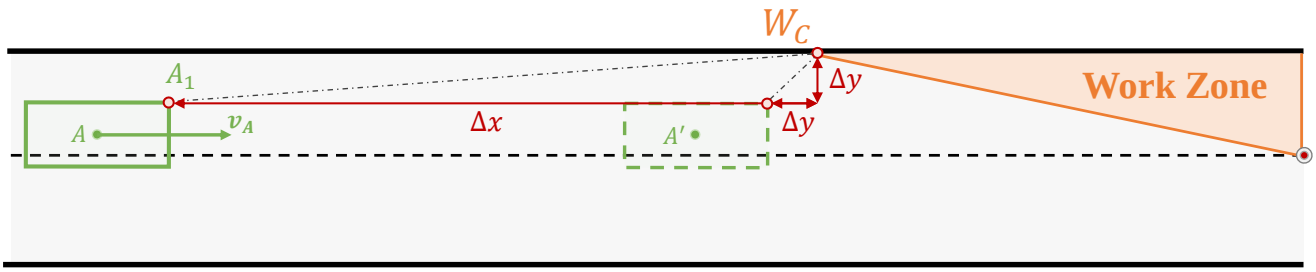


Figure S4: Geometric illustration of ACT for a vehicle approaching the work-zone boundary.

and begins its lateral adjustment, with CEDI approaching zero near $t = 24.0$ s.

Consistent with Case 1 in the main article, during the SV's aborted overtaking attempt, CEDI provides earlier warning of elevated risk than the comparison SSMs. This information can support timely decisions to abandon overtaking and select a safer merging opportunity.

Additional case S2: A vehicle conflict under surrounding traffic constraints

Fig. S3 illustrates a vehicle conflict in which the faster SV approaches a slower interacting vehicle. CEDI captures different risk evolutions for the two vehicles: the SV's CEDI peaks around $t = 389.6$ s and then remains relatively stable as it continues to decelerate, whereas the interacting vehicle's CEDI reaches a higher peak around $t = 391.2$ s.

During this interval, the SV's braking reduces the closing speed, but the longitudinal gap continues to narrow. The SV can mitigate the conflict through continued braking, while the interacting vehicle faces more restricted options: further braking may intensify the conflict with the approaching SV, and steering may introduce conflicts with surrounding vehicles 1 and 2. The continued gap reduction, together with these constraints, is consistent with the interacting vehicle's later and higher CEDI peak. As the closing speed decreases further, both CEDI values decline and approach zero near $t = 393.9$ s.

Compared with the other SSMs, CEDI captures the risk asymmetry between the interacting vehicles in this case. It reflects not only the temporal proximity of a potential collision but also how the vehicles' actual evasive actions change risk during the interaction.

S6. Explanation of abrupt changes in ACT for MLC conflicts

ACT characterizes the approach between interacting objects using their shortest distance and closing rate (Venthuruthiyil and Chunchu, 2022). Let δ denote the shortest distance between the interacting objects. For approaching motion, define the closing rate as $c = -\partial\delta/\partial t > 0$. ACT is then

$$ACT = \frac{\delta}{c} = -\frac{\delta}{\partial\delta/\partial t}. \tag{S10}$$

Let W_s denote the upstream starting point of the work zone and W_c the closest point on its boundary to vehicle A (**Fig. S4**). With the work zone treated as stationary and vehicle A assumed to maintain a constant velocity, ACT can be expressed as

$$ACT = \frac{\left\|\overrightarrow{A_1 W_c}\right\|^2}{\boldsymbol{v}_A \cdot \overrightarrow{A_1 W_c}} \tag{S11}$$

Let Δx and Δy denote the longitudinal and lateral components of $\overrightarrow{A_1 W_c}$, respectively. Expanding the geometric expression gives

$$\begin{aligned} ACT &= \frac{\left\|\overrightarrow{A_1 W_c}\right\|^2}{\boldsymbol{v}_A \cdot \overrightarrow{A_1 W_c}} \\ &= \frac{(\Delta x)^2 + (\Delta y)^2}{v_A\left(\Delta x \cos\theta_A + \Delta y \sin\theta_A\right)}. \end{aligned} \tag{S12}$$

Assuming that vehicle A travels parallel to the road's longitudinal direction, $\sin\theta_A = 0$ and $\cos\theta_A = 1$. While the upstream endpoint W_s remains the closest point on the work-zone boundary, ACT simplifies to

$$ACT = \frac{(\Delta x)^2 + (\Delta y)^2}{v_A \Delta x}, \qquad \Delta x > 0, \tag{S13}$$

where Δx and Δy denote the longitudinal and lateral displacements from vehicle A to W_s, respectively.

Under constant-velocity motion, Δy remains constant, whereas Δx decreases as vehicle A moves forward. Differentiating ACT with respect to Δx yields

$$\frac{\partial ACT}{\partial \Delta x} = \frac{1}{v_A}\left(1 - \frac{(\Delta y)^2}{(\Delta x)^2}\right). \tag{S14}$$

For $\Delta y \neq 0$, setting this derivative to zero gives $\Delta x = |\Delta y|$. Thus, ACT reaches its minimum in this configuration when the longitudinal distance to W_s equals the lateral distance. Denoting this instant by t_m, the corresponding minimum is

$$ACT(t_m) = \frac{2|\Delta y|}{v_A}. \tag{S15}$$

This observation does not undermine the underlying rationale of ACT, but highlights the need to consider work zone geometry when applying it to MLC conflicts under the CV assumption.

References

Venthuruthiyil, S.P., Chunchu, M., 2022. Anticipated collision time (ACT): A two-dimensional surrogate safety indicator for trajectory-based proactive safety assessment. Transportation Research Part C: Emerging Technologies 139, 103655. doi: 10.1016/j.trc.2022.103655.